

Programme	B. Sc. Computer Science				
Course Title	Introduction to Digital Marketing				
Type of Course	SEC				
Semester	V				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	3	3	-		45
Pre-requisites	1. Basic Computer Literacy 2. Familiarity with Online Platforms 3. Willingness to Learn				
Course Summary	Introduction to Digital Marketing" provides students with a foundational understanding of key concepts and techniques in the rapidly evolving field of digital marketing. Through engaging lectures. Students will explore various digital marketing channels, including search engine optimization (SEO), social media marketing, email marketing, and content marketing				

Course Code		Course Title Introduction to Digital Marketing		
Credit 3		Duration 45 hrs		
Sl. NO:	Course Outcome	Cognitive level *	Knowledge category #	Evaluation Tools used
CO1	To understand the concept of digital marketing and its integration with traditional marketing	U	C	Instructor-Create Exams or Quiz
CO2	To understand customer value journey in digital context and behaviour of online consumers	A	P	Discussions and Quizzes

CO3	To examine various tactics for enhancing a website's position and ranking with search engines	U	F	Instructor created exams or Home assignments
CO4	To Identify and differentiate between various digital marketing channels, including SEO, social media, email, and content marketing.	A ,E	P	Discussions, Quizzes
CO5	To get overall idea in implementing basic digital marketing strategies to enhance online visibility and engagement.	Ap	P	Viva Voce Observation of practical skills
CO6	To get to know about ethical considerations and best practices in digital marketing, including privacy, data protection, and consumer trust	U	M	Instructor Created - Exams, Assignments
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus

Module	Unit	Content	Hrs	Marks
I	Digital Marketing Basics		9	12
	1	Overview of digital marketing	2	
	2	Importance of digital marketing for businesses	2	
	3	Introduction to key digital marketing channels (SEO, social media, email marketing)	3	
	4	Basics of creating a digital marketing strategy	2	
II	Content Marketing & Social Media		9	12
	5	Content Marketing Fundamentals	2	
	6	Content Strategy Development	2	
	7	Content Creation for Different Platforms	2	
	8	Introduction to Social Media Marketing & keyword Optimization	2	
	9	Social Media Strategy & Community Management	1	
III	Search Engine Optimization (SEO) & Paid Advertising		9	14
	10	Introduction to Search Engine Optimization	2	

	11	On-page and Off-page SEO Techniques	2	
	12	Search Engine Marketing (SEM) Fundamentals	2	
	13	Pay-Per-Click (PPC) Advertising with Google Ads	2	
	14	Social Media Advertising Platforms	1	
IV	Web Analytics & Emerging Trends		9	12
	13	Introduction to Web Analytics & Key Metrics	2	
	14	Using Analytics Tools for Data-Driven Decision Making	2	
	15	Conversion Tracking & Optimization	2	
	16	Emerging Trends in Digital Marketing	2	
	17	The Future of Marketing	1	
V	Open Ended Module Practical Applications, Case Study and Course Project			
	1	Social Media Marketing-Social media Channels	2	
	2	Leveraging social media for brand conversions and buzz	2	
	3	Recent trends in digital marketing	3	
	4	Demonstrate how to use google web masters Indexing Using API	2	

References:

1. Digital Marketing for Dummies by Ryan DeWald
2. MARKETING 4.0 Moving from Traditional to Digital PHILIP KOTLER
HERMAWAN KARTAJAYA IWAN SETIAWAN
3. Ryan, D. (2014). Understanding Digital Marketing: Marketing Strategies for Engaging the Digital Generation, Kogan Page Limited
4. Taxmanns - Digital Marketing - Satinder Kumar, Supreet Kaur
5. Social Media Marketing 2024 - Mastering New Trends & Strategies for Online Success - Robert Hill

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Practical Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓	✓		✓
CO 3	✓			✓
CO 4	✓			✓
CO 5		✓		✓
CO6				✓

Model Question Papers/ Major

FIRST SEMESTER (CUFYUGP) DEGREE EXAMINATIONS, OCTOBER 2024

Computer Science

CSC1CJ101 - Fundamentals of Computers and Computational Thinking

(2024 Admissions)

Time: Two Hours

Maximum: 70 Marks

Section A

[Answer All. Each question carries 3 marks] (Ceiling 24 Marks)

1. Briefly describe the historical development of computers mentioning two key figures and their contributions.
2. Explain the concept of the Von Neumann architecture.
3. Convert the following numbers from decimal to binary: (a) 25, (b) 100.
4. Differentiate between active and passive electronic components. Provide an example of each.
5. What is the function of a motherboard? List four key components on a motherboard.
6. Distinguish between application software and system software. Give an example of each.
7. Briefly explain the role of an operating system in a computer system.
8. Define the term "computational thinking."
9. What are the steps involved in problem decomposition?
10. Explain the difference between inductive and deductive reasoning.

Section B

[Answer All. Each question carries 6 marks] (Ceiling 36 Marks)

11. Describe the evolution of computers from first generation to present day, highlighting the key features of each generation.
12. Explain the concept of digital codes with reference to Gray code and BCD.
13. Briefly explain the working principle of a transistor.
14. Differentiate between RAM and ROM. Explain the different types of RAM.
15. Discuss the different types of operating systems and their characteristics.
16. Explain the concept of booting with reference to POST and UEFI/Legacy BIOS.
17. Describe the four key pillars of computational thinking.
18. Explain the importance of algorithms in solving problems. Discuss the qualities of a good algorithm.

Section C

[Answer any one. Each question carries 10 marks] (1 x 1 = 10 Marks)

19. a) Discuss the contributions of John von Neumann to the field of computing.
- b) Briefly explain the concept of Quantum Processing Units (QPU) and their potential applications.
20. a) Describe the various components of a computer system and their functionalities.
- b) Explain the need for device drivers in a computer system.

FIRST SEMESTER (CUFYUGP) DEGREE EXAMINATIONS, OCTOBER 2024

Computer Science

CSC1FM105 - Data Analysis and Visualization using Spreadsheets

(2024 Admissions)

Time: 1.5 Hours

Maximum: 50 Marks

Section A

[Answer All. Each question carries 2 marks] (Ceiling 16 Marks)

1. Define the terms "worksheet" and "workbook" in the context of Excel.
2. Explain three formatting options available for cells in Excel and briefly describe their applications.
3. How can you insert a new row and a new column within an Excel sheet?
4. Write a formula to calculate the average of a range of cells (A1:A10) in Excel.
5. Explain the purpose and benefits of data validation in Excel.
6. Differentiate between absolute and relative cell referencing with an example for each.
7. Describe the concept of Autofill and give an example of how it can be used in Excel.
8. Explain the steps involved in filtering data based on a specific criterion in Excel.
9. Describe the process of importing data from a text file into an Excel spreadsheet.
10. List two commonly used functions for applying conditional formatting in Excel.

Section B

[Answer All. Each question carries 6 marks] (Ceiling 24 Marks)

11. A dataset contains duplicate entries. Describe the steps involved in removing these duplicate rows in Excel.
12. You are given a dataset with sales figures for different regions. Explain how you would create a pivot table to analyse trends in sales across these regions.
13. Explain the concept of "What-If Analysis" in Excel and provide an example of how it can be used to support decision-making.
14. Write a formula using the VLOOKUP function to find the product price based on a product code in another table.
15. Describe three different chart types suitable for visualizing data in Excel and explain when you might use each type.

Section C

[Answer any one. Each question carries 10 marks] (1 x 1 = 10 Marks)

16. You are provided with a large dataset containing customer information and sales data.
- Describe how you would utilize advanced features like data filtering, sorting, and pivot tables to identify the top 5 customers by sales in a specific region for the past year.
 - Create a visually appealing dashboard in Excel that summarizes key customer and sales data, including a chart to represent the top-selling products.

(or)

17. Explain the concept of macros in Excel and discuss their potential benefits and drawbacks. Briefly describe the steps involved in creating a simple macro to automate a repetitive task.
